www.tntextbooks.in

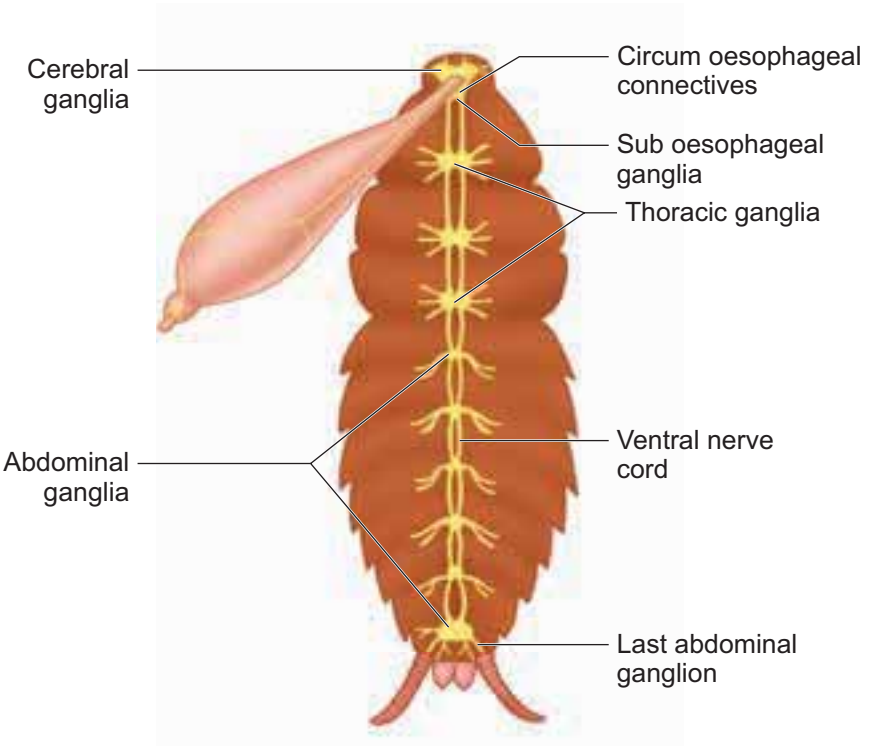


Figure 4.13 *Periplaneta americana*: Nervous system

2000 simple eyes called the **ommatidia** (singular: *ommatidium*), through which the cockroach can receive several images of an object. This kind of vision is known as mosaic vision with more sensitivity but less resolution.

Arthropod eyes are called compound eyes because they are made up of repeating units, the ommatidia, each of which functions as a separate visual receptor.

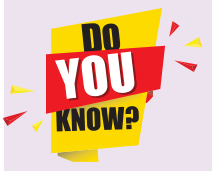
What is the difference between compound eyes and simple eyes?

Why is mosaic vision with less resolution seen in cockroaches?

Excretory system

The Malpighian tubules are the main excretory organs of cockroach which help in eliminating the nitrogenous wastes from the body in the form of uric acid. Cockroach excretes uric acid, so it is **uricotelic**. In addition, fat body, nephrocytes, cuticle, and urecose glands are also excretory in function.

The malpighian tubules are thin, long, filamentous, yellow coloured structures attached at the junction of midgut and hindgut. These are about 100-150 in number



Marcello Malpighi – described these tubules and called them vasa varicose. Meckel later called them Malpighian tubules.